SONiX 32-Bit Cortex-M0 Micro-Controller

DMA ADC Example Code

AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/27	1. First version.

Table of Content

1 OVERVIEW

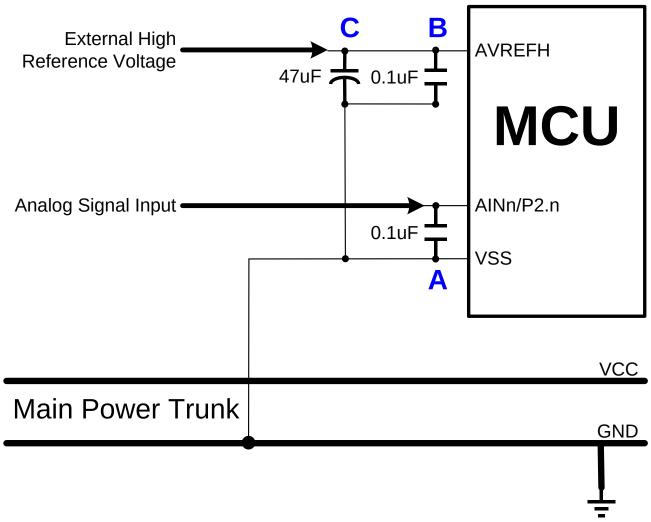
This document provides our customers the C program examples of ADC in DMA mode.

2 FEATURE

- Analog to Digital Convertor function with DMA transfer.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- The ADC input signal must be through a 0.1uF capacitor. The 0.1uF capacitor is set between ADC input pin and VSS pin, and must be on the side of the ADC input pin as possible. Don't connect the capacitor's ground pin to ground plain directly, and must be through VSS pin. The capacitor can reduce the power noise effective coupled with the analog signal.



4 EXAMPLE CODE

4.1 Initialization

Initial ADC and DMA function. User can modify the settings of ADC and DMA in these function.

```
// Initial ADC
ADC_FuncInit(ADC_DIV32, ADC_12BIT, Multiple_Channel, Continuous_Mode);

// Init DMA
DMAADCInit();
```

For example: Set ADC high reference voltage source as Internal VDD.

ADC resolution as 12 bit.

ADC conversion time = 1 / (ADC MCLK/32) * 16 sec.

```
// Initial ADC
ADC_FuncInit(ADC_DIV32, ADC_12BIT, Multiple_Channel, Continuous_Mode);
```

```
void ADC_FuncInit(uint8_t bPCLKDiv, uint8_t bADCLen, uint8_t bCHMode, uint8_t bSCMode)
{
  __ADC_ENABLE_HCLK;                                //Enables HCLK for ADC

  SN_ADC->ADM_b.AVREFHSEL = ADC_AVREFHSEL_INTERNAL;  //Set ADC high reference voltage source from internal reference
  SN_ADC->ADM_b.VHS = ADC_VHS_INTERNAL_4P5V;         //Set ADC high reference voltage source as Internal 4.5V

  SN_ADC->ADM_b.GCHS = ADC_GCHS_EN;                  //Enable ADC global channel

  SN_ADC->ADM_b.ADLEN = bADCLen;                     //Set ADC resolution = 12-bit

  SN_ADC->ADM_b.ADCKS = bPCLKDiv;                    //ADC_CLK = ADC_PCLK/32

  SN_ADC->CONVCTRL_b.SCMODE = bSCMode;               //Set mode

  SN_ADC->CONVCTRL_b.CH = bCHMode;                   //Set converting channel

  SN_ADC->ADM_b.ADENB = ADC_ADENB_EN;                //Enable ADC

  UT_DelayNx10us(10);                                //Delay 100us

  SN_ADC->ADM1_b.ACS = ENABLE;                       //Calibration start
  while (SN_ADC->ADM1_b.ACS == 1);

  SN_ADC->ADM1_b.CALIVALENB = ENABLE;                //ADC conversion with calibration value
}
```

The setting of DMA should refer to the table in the datasheet.

```
void DMAADCInit(void)
{
  DMA_InitSt stDMACH_Init;

  // Set DMA CH0 as P(ADC) to M(SRAM) mode
  // Source : Peripheral ADC, Fixed, 8bit, ADC, Burst 4
  stDMACH_Init.b_SrcAddrCtrl = DMA_SRCAD_CTL_FIX;
  stDMACH_Init.b_SrcMode     = DMA_SRC_PERIPHERAL;
  stDMACH_Init.b_SrcWidth    = DMA_SRC_WIDTH_16BIT;
  stDMACH_Init.b_SrcReqSel   = DMA_SRCRS_ADC;
  stDMACH_Init.b_BurstSize   = DMA_BURST_4;
  // Destination : Memory, Increment, 8bit, no request
  stDMACH_Init.b_DstAddrCtrl = DMA_DSTAD_CTL_INC;
  stDMACH_Init.b_DstMode     = DMA_DST_MEMORY;
  stDMACH_Init.b_DstWidth    = DMA_DST_WIDTH_16BIT;
  stDMACH_Init.b_DstReqSel   = DMA_DSTRS_NONE;
  // Channel : LV0, Enable TC/ABT interrupt
  stDMACH_Init.b_Priority    = DMA_CHPRI_LV0;
  stDMACH_Init.b_IntTCEn     = DMA_INT_TC_MSK_DIS;
  stDMACH_Init.b_IntABTEn    = DMA_INT_ABT_MSK_DIS;

  // Initial DMA
  DMA_Init(&stDMACH_Init, eDMA_CH0);
}
```

4.2 DMA ADC function enable

DMA request will be issued when the ADC FIFO is greater than the threshold. The DMA TC flag and interrupt will be set when all data transfers are completed.

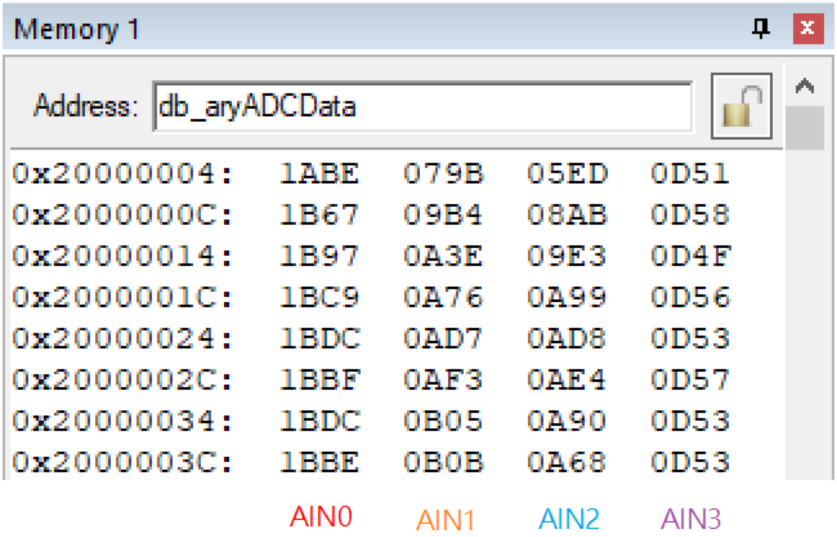
```
  // Start DMA ADC
  DMAADCStart();
  ADC_DMA_Start(ADC_CHS_AIN0|ADC_CHS_AIN1|ADC_CHS_AIN2|ADC_CHS_AIN3, ADC_DMA_FIFO_TH_3, DMA_TEST_SIZE);

  // Wait for DMA TC interrupt flag
  while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);

  while(1); // Pass
}
```

5 RESULT

The ADC conversion result will be moved to SRAM by DMA. The results can be observed through Watch window or Memory window.



Main Office:
Address: 10F-1, NO. 36, Taiyuan Street.,Chupei City, Hsinchu, Taiwan R.O.C.
Tel: 886-3-5600 888
Fax: 886-3-5600 889
Taipei Office:
Address: 15F-2, NO. 171, Song Ted Road, Taipei, Taiwan R.O.C.
Tel: 886-2-2759 1980
Fax: 886-2-2759 8180
Hong Kong Office:
Unit No.705,Level 7 Tower 1,Grand Central Plaza 138 Shatin Rural Committee Road, Shatin, New Territories, Hong Kong.
Tel: 852-2723-8086
Fax: 852-2723-9179
Technical Support by Email:
Sn8fae@sonix.com.tw